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A Multidimensional Psychometric Scale for Measuring AI Dependency in Academic Research

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ABSTRACT

As artificial intelligence (AI) becomes embedded in academic workflows, concerns are increasing about researchers developing psychological and behavioral dependence on AI systems. Although AI tools can significantly enhance research efficiency and analytical capabilities, excessive or uncritical reliance may lead to cognitive disengagement, emotional over attachment, and maladaptive academic practices. Yet, the literature still lacks a validated instrument that can systematically assess AI dependency within scholarly contexts. This study addresses this gap by developing a multidimensional scale that draws on the affect, behavior, and cognition (ABC) model as the foundational structure of psychological response, incorporates basic psychological needs theory (BPN) to explain motivational dynamics, and is ultimately grounded in the I-PACE model, which provides a comprehensive framework for understanding the development of dependency-related behaviors. The scale was developed using a rigorous mixed-methods design implemented across a five-stage sequence that encompassed systematic item generation, expert review for content adequacy, and extensive empirical evaluation with a large academic sample. Findings from exploratory and confirmatory factor analyses converged to support a stable three-factor structure consisting of fifteen theoretically coherent items. The instrument demonstrated strong psychometric integrity, including high internal consistency, robust convergent and discriminant validity, and meaningful predictive validity, particularly in relation to attitudes linked to academic research misconduct. By establishing a validated measure of AI dependency within research contexts, this study advances current debates in digital ethics and responsible AI use and offers an empirically grounded tool for strengthening research integrity practices in higher education.

KEYWORDS

AI dependency; academic integrity; cognitive; emotional; behavioral reliance

1. Introduction

The rapid rise of artificial intelligence (AI) as a transformative technological ecosystem is reshaping multiple sectors, including education, research, and scholarly communication (Fui-Hoon Nah et al., 2023; Lund et al., 2023). Across higher education, AI systems ranging from intelligent writing assistants and search-enhancement tools to automated analytics, recommendation engines, and adaptive research-support platforms are increasingly embedded in everyday academic practice (Hanafi et al., 2025). These systems provide scholars with real-time support for information processing, writing, editing, pattern recognition, decision support, and workflow automation, thereby enhancing productivity, efficiency, and academic output (Zhang et al., 2025; Al-Zahrani, 2024). As AI gradually integrates into nearly every stage of the research lifecycle, from literature synthesis and ideation to data analysis and manuscript preparation, it has become indispensable for many researchers. However, this growing integration has also raised significant psychological, ethical, and methodological concerns related to scholars' potential overreliance on AI technologies (Ali et al., 2024).

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The expanding use of AI tools, whether generative, predictive, assistive, or analytic, is increasingly shaped by users' trust in these systems (Klingbeil et al., 2024). As trust deepens, researchers may shift from using AI as a supplementary aid to depending on it as a primary source of interpretation, problem-solving, and evaluative judgment (Wang et al., 2025). Such shifts alter the nature of scholarly inquiry by potentially displacing core academic competencies such as critical thinking, ethical reasoning, methodological rigor, and independent intellectual effort (Mishra & Anthony, 2024). A growing body of literature cautions that researchers may uncritically incorporate AI-generated or AI-mediated information into their work, weakening the rigor of scholarly evaluation (Jacovi et al., 2021; Kotsis, 2024). Furthermore, the increasing automation of text, ideas, and analytic reasoning raises questions about authorship transparency, intellectual accountability, and research integrity, especially when AI contributions are not rigorously validated or clearly attributed (Colella, 2025; Dwivedi et al., 2023). Overreliance on AI, whether for writing, decision support, or analytic tasks, has therefore been linked to plagiarism, intellectual disengagement, and ambiguous authorship practices (Désiron & Petko, 2023; Shengelia et al., 2024; Zunaidah et al., 2023). As several scholars caution, the ease with which AI systems generate or shape publishable content poses significant risks to academic integrity if left unregulated, particularly when AI-mediated outputs are incorporated without rigorous validation or critical appraisal (Ogwueleka, 2025).

While these cognitive and ethical concerns are substantial, they represent only part of a broader and more complex pattern of psychological dependency on AI (Wang et al., 2023). Emerging studies show that AI tools can also fulfill emotional and behavioral functions for researchers reducing anxiety, providing reassurance, and supporting habitual academic routines (Gillath et al., 2021; Zhang et al., 2024). This suggests that AI technologies are not only functional aids but also affective and behavioral companions that shape researchers' emotional regulation and work engagement (Igbokwe, 2023). Understanding such layered psychological reliance is essential for ensuring responsible and balanced integration of AI within scholarly environments.

Despite growing interest in AI-related behaviors, the field lacks a validated psychometric instrument that captures the cognitive, emotional, and behavioral dimensions of AI dependency specifically in academic research contexts. Existing measures either focus narrowly on students, general technology overuse, or classroom-based AI tools, and do not reflect the unique psychological dynamics associated with research-oriented AI engagement. To address this gap, the present study introduces and validates the AI Dependency Scale for research scholars, a theory-driven instrument designed to assess multidimensional psychological reliance on AI technologies within research practice. In addition, the study examines whether such dependency is associated with vulnerabilities in academic integrity, particularly plagiarism-related behavior.

Accordingly, this study is guided by the following research questions:

RQ1: To what extent does the proposed AI Dependency Scale reliably and validly measure cognitive, emotional, and behavioral reliance on AI tools among academic researchers?

RQ2: How is AI dependency associated with academic dishonesty, particularly the attitude of research scholars toward plagiarism?

2. Literature review

The integration of artificial intelligence (AI) technologies into academic research has evolved far beyond their initial role as supplementary aids. A wide range of AI systems, including generative tools such as ChatGPT, Scite.ai, and GrammarlyGO, as well as predictive, assistive, and analytic algorithms, are increasingly embedded in higher education and scholarly practice (Bacallo et al., 2024). Unlike earlier educational technologies designed primarily for information access or content delivery, contemporary AI systems automate routine academic tasks, enhance productivity, and support complex functions such as summarization, data interpretation, paraphrasing, pattern recognition, and reference generation (Abbas et al., 2023; Lo, 2023). By simulating or augmenting human-like reasoning, these AI tools are reshaping how researchers engage with intellectual labor (de Winter et al., 2025) and moving AI from a peripheral support mechanism to a core element of academic workflows.

As AI technologies become more deeply embedded in daily academic routines, emerging research highlights a growing pattern of psychological and behavioral dependency (Naseer et al., 2025). Importantly, dependency represents a construct distinct from general adoption or technological proficiency (Saihi et al., 2025). Traditional technology adoption frameworks such as the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) emphasize perceived usefulness and intentions to use, whereas dependency involves deeper psychological attachment and the tendency to rely on AI for tasks that would otherwise require human judgment, cognitive effort, or reflective reasoning. This distinction aligns with media dependency theory, which suggests that individuals increasingly rely on communication technologies when these tools help fulfill personal, professional, or societal goals (Ball-Rokeach & DeFleur, 1976). In academic contexts, AI systems, whether generative or analytic, may become indispensable not only for improving efficiency but also for offering cognitive scaffolding, emotional reassurance, and habitual ease. This mechanism mirrors patterns observed in other digital dependency domains, including internet addiction (Young, 1998), smartphone addiction (Kwon et al., 2013), and social media overuse (Bányai et al., 2017). These literatures collectively support the argument that AI-driven tools are uniquely positioned to foster maladaptive reliance among researchers operating under competitive and high-performance pressures.

Growing ethical and psychological concerns further underscore the risks of overreliance on AI systems. Jacovi et al. (2021) warn that uncritical trust in AI outputs, particularly those generated through generative models may diminish users' capacity to evaluate information critically. Other studies similarly link excessive dependence on AI tools to reduced intellectual autonomy, weakened critical thinking, and an increased likelihood of academic dishonesty (Capinding, 2024; Salah et al., 2024). Students and researchers experiencing stress or workload pressure often use AI systems to offload cognitive demands or obtain emotional comfort, which over time can develop into habitual and deeply ingrained usage patterns (Zhang et al., 2024). Gillath et al. (2021) likewise describe how AI tools may serve as emotional companions, offering affirmation or a sense of control during demanding academic tasks. These behaviors parallel well-documented patterns in the cognitive offloading literature, where individuals increasingly rely on external tools to support memory, decision-making, and problem-solving (Risko & Gilbert, 2016).

Recent scholarship has paid particular attention to the emotional and cognitive dimensions of AI engagement. Studies show that researchers and students may form affective attachments to AI systems, especially those capable of conversational or personalized interaction (Gillath et al., 2021; Naseer et al., 2025). For instance, Zhang et al. (2024) found that unmet psychological needs and academic stress are strong predictors of AI overuse, while Ji et al. (2025) identified emotional exhaustion as a significant correlate of heightened AI reliance. These findings suggest that AI is increasingly used not merely for functional purposes but also as a mechanism for emotional regulation and academic reassurance. Emotional dependence may manifest through confidence-seeking behaviors, while cognitive reliance manifests through the habitual use of AI for ideation, interpretation, or writing. Such reliance raises significant ethical concerns, as scholars have observed connections between AI dependency and academically questionable practices such as plagiarism, authorship misrepresentation, and unacknowledged use of AI-generated or AI-mediated content (Désiron & Petko, 2023; Shengelia et al., 2024). These patterns are consistent with broader concerns about academic dishonesty in digital environments (King, 2023), reinforcing the need to understand the ethical implications of AI dependency.

Concerns related to academic integrity have become particularly salient as AI systems expand their capabilities. Whether through generative text, automated reasoning, or predictive analytics, AI-mediated outputs can blur the line between human work and technological assistance. Excessive reliance on AI has been associated with plagiarism, diminished personal accountability, and weakened ethical judgment (Dwivedi et al., 2023; O'Connor, 2024). Critics such as Chomsky, as cited by Khalaf (2025), describe some AI applications as "high-tech plagiarism," underscoring the risk of bypassing essential learning and research processes. Issues surrounding intellectual property, fabricated outputs, and the production of convincing but unverified content further challenge scholarly credibility and policy discourse (Else, 2023; Lund et al., 2023). These ethical concerns justify examining plagiarism-related outcomes within the nomological network of AI dependency, as dependence, whether cognitive or

behavioral may encourage shortcut practices, uncritical content appropriation, and reduced scholarly ownership.

To measure these emerging concerns, several instruments have been developed to assess AI dependence or technology overuse. Morales-García et al. (2024) introduced the Dependence on Artificial Intelligence (DAI) Scale to capture general reliance on AI among university students, while Jorolan et al. (2025) developed the MAD-AI scale to examine dependency among educators. Capinding (2024), by contrast, proposed a task-based typology of AI reliance focused on academic reading, writing, and arithmetic. However, despite these contributions, existing measures tend to be unidimensional, population-specific, or limited to classroom settings. Notably, no validated instrument currently captures the multidimensional psychological nature of AI dependency among academic researchers a group uniquely responsible for producing and disseminating original knowledge.

To address this gap, the present study develops and validates a psychometric instrument designed to measure AI dependency among academic researchers. Anchored in the ABC model of attitude structure (Eagly & Chaiken, 1993), the study conceptualizes dependency as comprising three interrelated dimensions: cognitive, emotional, and behavioral reliance. These dimensions are elaborated upon in the subsequent theoretical framework. In alignment with best practices for scale development, the study follows established guidelines for item generation, expert evaluation, and psychometric validation. Seminal guidance from DeVellis (2016), Hinkin (1998), and Boateng et al. (2018) emphasizes grounding scale construction in theory, followed by empirical testing through rigorous factor analysis. Accordingly, this study aims to produce a theoretically sound, psychometrically robust, and contextually relevant AI Dependency Scale tailored to academic research environments.

3. Theoretical background

The increasing dependency on AI tools within academia presents not merely a technological shift but a psychological transformation in how individuals engage with knowledge, problem-solving, and scholarly practices. To better understand this emerging phenomenon, this study synthesizes insights from the Interaction of Person–Affect–Cognition–Execution (I-PACE) model and Basic Psychological Needs Theory (BPN) to explain the underlying drivers of AI dependency. In parallel, the ABC model of attitude formation (Affect–Behavior–Cognition) is adopted to conceptualize the structure of dependency and guide the development of its multidimensional measurement.

The I-PACE model (Brand et al., 2025) provides a robust framework for analyzing the development of problematic or compulsive technology use. It posits that personal traits (e.g., impulsivity, perfectionism), affective responses (e.g., anxiety, frustration), and cognitive biases (e.g., overestimating performance expectations) interact dynamically, leading to maladaptive behaviors such as compulsive use or dependency. This model has been effectively applied to digital media and internet addictions, and recent extensions have supported its relevance in explaining AI overuse (Zhang et al., 2024). In the context of academic research, scholars often operate under conditions of sustained academic pressure, emotional fatigue, and performance anxiety, factors widely recognized as integral to the research environment. In such contexts, GenAI tools may serve as accessible supports for managing stress, reducing cognitive effort, or navigating research tasks more efficiently. Over time, these coping mechanisms may evolve into patterns of psychological dependency, particularly when reliance on AI substitutes independent academic engagement.

Complementing the I-PACE model, the Basic Psychological Needs Theory (Deci & Ryan, 2000) suggests that the frustration of fundamental needs autonomy, competence, and relatedness can lead individuals to adopt external compensatory mechanisms, such as AI technologies. When researchers feel inadequately skilled (competence), constrained in their decision-making (autonomy), or disconnected from academic communities (relatedness), they may gravitate toward AI tools as a means of restoring psychological balance. Studies have shown that students and researchers often turn to AI as a way to quickly obtain feedback, manage academic workloads, and gain a sense of accomplishment, particularly in the face of these motivational deficits (Capinding, 2024; Morales-García et al., 2024). While these tools offer short-term relief, long-term reliance may reinforce passive academic engagement and hinder independent learning and originality.

While the I-PACE and BPN theories offer explanatory depth, the ABC model of attitude (Eagly & Chaiken, 1993) serves as a useful structural lens to delineate the core dimensions of dependency. According to this model, affect represents emotional reliance, using AI for stress relief, motivation, or emotional comfort. Behavior refers to habitual or compulsive actions repeatedly using AI, even when unnecessary. Cognition encompasses belief systems, including perceptions of AI as a superior or indispensable academic aid. In this study, the ABC framework serves as the organizing principle for developing the AI Dependency Scale, capturing the multifaceted nature of reliance through measurable indicators across these three domains. However, the model is not employed as a causal framework, but rather as a descriptive taxonomy of the psychological and behavioral manifestations of AI dependency.

By integrating these theoretical perspectives, this study addresses both the *why* (motivational and emotional mechanisms) and the *how* (multidimensional structure) of AI dependency. The I-PACE and BPN theories ground the study in psychological science, offering insight into the processes that drive overuse, while the ABC model provides the operational lens for scale development and measurement of the construct. This theoretical integration allows for a comprehensive understanding of AI dependency that is both psychologically explanatory and empirically measurable, making it well-suited for application in higher education contexts where ethical, cognitive, and emotional stakes are high. Based on the synthesis of these theoretical insights, this study conceptualizes AI dependency as a multidimensional construct consisting of three interrelated components: cognitive reliance (dependence on AI for intellectual tasks), emotional reliance (use of AI for stress relief or reassurance), and behavioral reliance (habitual or compulsive use). This tripartite structure draws upon the ABC model of attitude (Eagly & Chaiken, 1993) and reflects the complex ways researchers engage with GenAI tools across affective, behavioral, and cognitive domains. Together, these theoretical perspectives provide a coherent foundation for conceptualizing and measuring AI dependency in this study.

4. Methodology

4.1. Study design

The dynamics of researchers' AI dependency are closely linked to their environments and the specific situations in which they work. Therefore, to better understand AI dependency, it is essential to consider the perspectives of scholars who utilize these technologies. Our methodology is grounded in this principle, employing mixed-method research techniques (qualitative & quantitative) to develop the AI dependency scale. We closely adhered to the established procedures suggested by Churchill (1979), and our approach is documented through five comprehensive studies that contribute to the systematic development and validation of this scale (Figure 1). In alignment with this multiphase design, the participant characteristics and data collection procedures are presented in the following subsection.

4.2. Participants and data collection procedure

Data collection for this study was organized into five sequential phases, with a clear methodological distinction between the initial qualitative phase and the subsequent quantitative phases. Phase 1 constituted an exclusively qualitative, developmental stage focused on scale construction. During this phase, qualitative data were generated through focus group discussions and a structured review of the literature to identify core indicators of AI dependency. These indicators were then subjected to expert-led face validation to refine and consolidate the preliminary item pool. Given the exploratory and instrument-development focus of Phase 1, no survey-based quantitative data were collected at this stage. In contrast, Phases 2 – 5 employed a quantitative research design.

Quantitative data were collected using offline, paper-based surveys administered across public, private, and deemed universities offering doctoral programs. In this study, public universities refer to government-funded institutions, private universities to self-financed institutions, and deemed universities to autonomous institutions formally recognized by the national regulatory authority. Inclusion of all three institutional categories was intended to capture diverse doctoral research environments and to strengthen the generalizability of the study findings. In the present study, public universities refer to

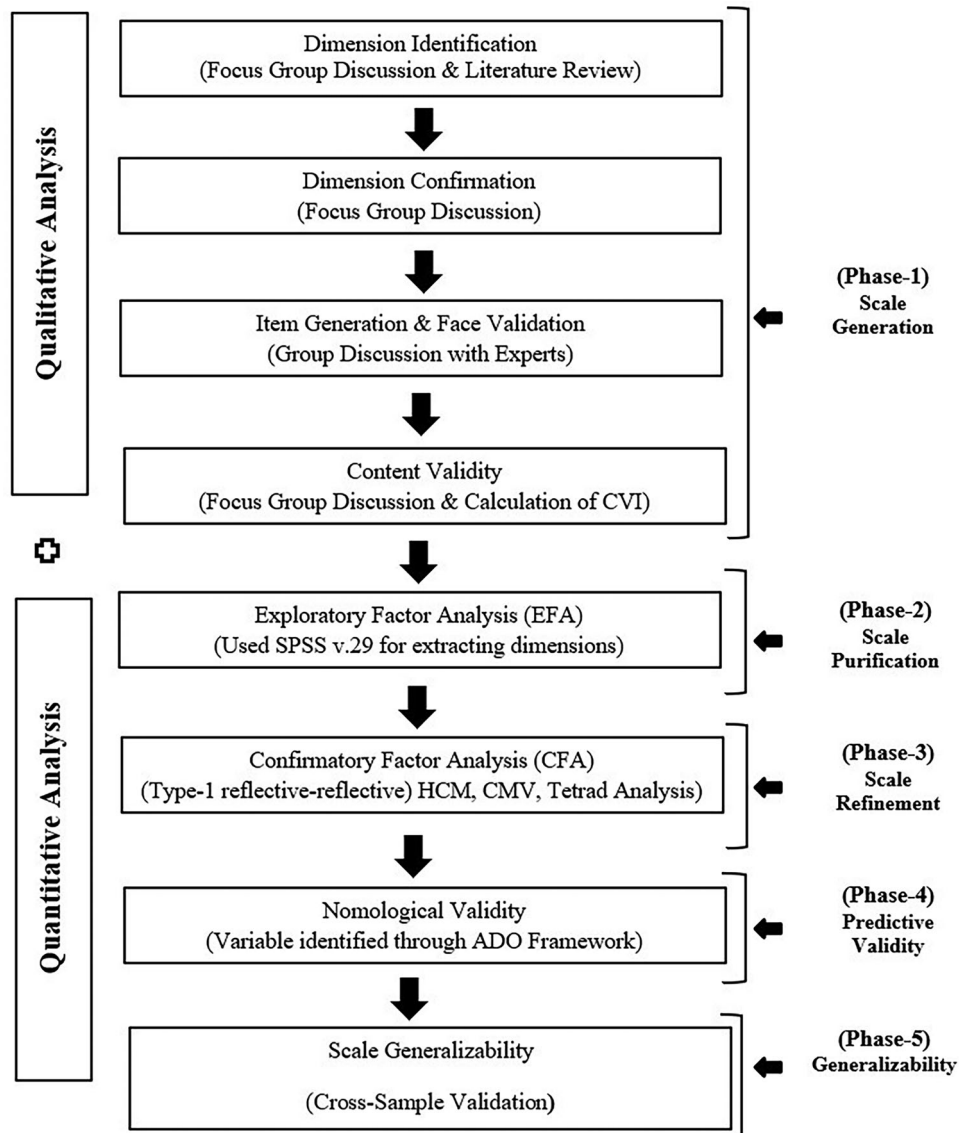


Figure 1. Steps followed for the development and validation of the AI dependency scale (Churchill, 1979).

government-funded institutions, private universities to self-financed institutions, and deemed universities to autonomous institutions formally recognized by the national regulatory authority; all three categories were included to ensure representation of diverse institutional contexts in doctoral research training and to enhance the generalizability of the findings.

The process of recruiting respondents began with approaching potential participants on-site at public, private, and deemed universities during survey administration. Prospective participants were provided with a brief explanation of the study's purpose and were invited to voluntarily complete the paper-based questionnaire. An informed consent statement was presented at the beginning of the survey, and only individuals who provided consent proceeded to participate. Participation was restricted to enrolled PhD or research scholars who were actively engaged in academic research, in accordance with the study's inclusion criteria. Quantitative data collection took place in early 2025 as part of a multi-phase validation process and adhered to institutional ethical guidelines.

Quantitative data were collected in four sequential phases (Phases 2–5), each involving an independent sample of research scholars. Collectively, these phases involved the recruitment of 1,550 scholars, of whom 1,255 provided complete and usable responses, yielding an overall response rate of 80.96%. A phase-wise breakdown of participation and response rates is provided: In Phase 2, 300 scholars were invited, and 248 usable responses were retained after data screening (response rate = 82.6%). In Phase

3, 260 scholars were approached, resulting in 219 valid responses (response rate = 84.2%). In Phase 4, 230 scholars were contacted, and 200 responses were retained after excluding incomplete questionnaires (response rate = 86.9%). In Phase 5, 760 scholars were invited, of whom 588 provided complete and valid responses, yielding a response rate of 76.3%. The detailed procedures and analyses conducted in each phase of the scale development process are described below.

4.2.1. Generation of items (Phase-1)

In phase 1, our objective was to create and develop a comprehensive set of statements to assess researchers' reliance on artificial intelligence (AI). This process involved a three-stage strategy: first, we conducted a focus group to gather initial insights on AI dependency; second, we reviewed existing literature to identify key aspects of AI dependency; and third, we organized a follow-up focus group to validate these identified aspects. Based on these activities, we developed an initial set of items and then conducted a face validity assessment. In order to confirm the validity of the created items, we utilized the content validity index method (Lynn, 1986; Polit & Beck, 2006). A group of 10 experts was gathered in a fourth focus group to assess the significance of each item to the concept. The items that didn't satisfy the predetermined criteria and standard were eliminated.

4.2.2. Scale purification (Phase-2)

This phase focused on refining the items developed in the initial stage. We conducted a preliminary study targeting PhD research scholars across various disciplines, including management, sciences, social sciences, and technology, in accordance with the 2014 education classification by the Department of Higher Education in India. Our selection comprised a diverse range of academic institutions listed by the University Grants Commission, which includes public, private, and deemed institutions. To enhance diversity and reduce sampling bias, we selected academic research scholars using a stratified purposive sampling technique based on institution type, discipline, and gender, and obtained informed consent prior to ultimately collecting data from 248 participants. We used exploratory factor analysis to streamline the list of items and evaluate the reliability of our measurements.

4.2.3. Scale refinement (Phase-3)

The third phase centered on the process of scale refinement. This involved 219 academic researchers, using sampling methods similar to those used in phase 2. Stratified sampling was maintained to preserve representational balance, and additional care was taken to avoid duplication across phases. The underlying factor structure was assessed through reflective Type 1 hierarchical component modeling. Next, we performed confirmatory factor analysis (CFA) to evaluate model fit, consistency of the construct, and both discriminant and convergent validity. Furthermore, we implemented Harman's single-factor test and marker variable analysis to mitigate potential common-method variance.

4.2.4. Predictive validity of AI dependency (Phase-4)

In this phase, we aimed to establish the validity of the AI dependency nomological network and its predictive capabilities regarding outcomes. Utilizing the Antecedents, Decisions, and Outcomes (ADO) framework developed by Paul and Benito (2018), we structured the variables and analyzed their interrelationships. The research specifically examined scholars' attitudes toward plagiarism as the dependent variable, gathering data from a sample of 200 scholars, consistent with the sampling techniques employed in earlier studies.

4.2.5. Cross-sample validation and generalizability of AI dependency scale (Phase-5)

In the final phase of the study, we examined the cross-sample stability and generalizability of the AI Dependency Scale using an independent validation sample of research scholars ($N=588$) drawn from diverse disciplinary backgrounds. This phase was designed to assess whether the factor structure established in the earlier phases could be reliably replicated in a separate and larger cohort representing a broad range of academic domains, including both theoretical and applied disciplines. Confirmatory

Table 1. Demographic information of the total sample considered for all the phases ($N = 1255$).

Demographic	Phase 2 ($n = 248$)		Phase 3 ($n = 219$)		Phase 4 (200)		Phase 5 ($n = 588$)	
	N	%	N	%	N	%	N	%
Gender								
Male	171	68.95	157	71.69	171	85.5	315	53.57
Female	77	31.05	62	28.31	29	14.5	273	46.42
Age								
22–25	25	10.08	14	6.39	10	5.00	50	8.50
26–30	80	32.26	90	41.10	88	44.0	198	33.67
31–35	74	29.84	59	26.94	57	28.0	200	34.01
36–40	40	16.13	36	16.44	30	15.0	90	15.30
Above 40	29	11.69	20	9.13	15	7.5	50	8.50
PhD tenure								
1st Year	100	40.32	84	38.35	105	52.5	116	19.72
2nd Year	60	24.19	46	21.00	37	18.5	106	18.02
3rd Year	40	16.13	36	16.44	14	7.00	190	32.31
4th Year	18	7.26	23	10.50	14	7.00	108	18.36
5th Year	30	12.09	30	13.69	30	15.0	68	11.56
Discipline								
Sciences	60	24.19	50	22.83	63	31.5	103	17.5
Social Sciences	50	20.16	45	20.55	32	16.0	159	27.04
Humanities and Arts	40	16.13	35	15.98	92	46	136	23.12
Engineering and Technology	70	28.23	60	27.40	9	4.5	120	20.40
Medical and Health Sciences	28	11.29	29	13.24	4	2.0	70	11.90

factor analysis (CFA) was employed to evaluate the stability and robustness of the previously established measurement model within this independent sample (Table 1).

5. Scale development and validation

5.1. Phase 1 scale generation

5.1.1. Dimension identification

The accuracy of measurements depends on a comprehensive understanding of the meaning and structure of undertaken constructs (Carpenter, 2018). To identify the dimensions associated with AI dependency, we used a combination of focus group discussions and a comprehensive literature review. Initially, the focus group aimed to gather key insights and generate keywords to ensure a thorough exploration of relevant aspects related to dependency. This session, which was held in January 2024, involved two postdoctoral scholars, four PhD candidates, and two research assistants. Following this, we conducted a literature review of articles published in reputed journals. The detailed dimensions identified in this review informed the development of our measurement scale (Figure 3).

To analyze the qualitative data collected from the initial and follow-up focus group discussions, we employed thematic analysis as outlined by Braun and Clarke (2006). All focus group sessions were audio-recorded with participants' consent and subsequently transcribed verbatim. Two independent coders systematically read the transcripts and performed open coding to identify recurring concepts related to AI dependency. Codes were then reviewed, categorized into themes, and mapped onto the three theoretical domains (cognitive, emotional, and behavioral). NVivo 12 software was used to organize the data and facilitate code comparison. To ensure coding consistency, we calculated inter-coder reliability using Cohen's Kappa, which yielded a coefficient of 0.82, indicating substantial agreement (Landis & Koch, 1977). Disagreements were resolved through consensus discussion. This rigorous qualitative process supported the initial generation and refinement of construct dimensions used in the item development phase.

5.1.2. Dimension confirmation

To validate our proposed model, we organized a second round of focus group discussions to ensure that all important aspects were included. This panel consisted of eight participants: two faculty members, three research associates, and three industry experts. They carefully evaluated the dimensions presented in Figure 2.

5.1.3. Item generation and face validation

Phase one focused on generating items that accurately represent the identified dimensions while filtering out those that did not clearly reflect the intended characteristics. We started with a preliminary set of 30 items. To ensure clarity and relevance, we held a third focus group discussion. This group assessed factors such as wording clarity, redundancy, and response formats (Netemeyer et al., 2003; Papadas et al., 2017). The panel consisted of eight members: three faculty members, two research associates, one industry specialist, and two experts in research methodology specializing in scale development. Based on their feedback, five items were eliminated, and rephrased three others for improved clarity. For example, the statement “I use AI tools daily in my research activities” was revised to “I use AI tools daily for my research activities.” As a result, we finalized 25 items that clearly represented the intended characteristics for subsequent content validity and assessment.

5.1.4. Content validity

The primary objective of this phase was to ensure that the developed items accurately reflect the concept being measured. Content validity confirms that the questionnaire comprehensively addresses the concept it is intended to evaluate (Clemmensen et al., 2021; Polit & Beck, 2006). In this study, we used the Content Validity Index (CVI), a widely recognized measure for assessing content validity (Kovacic, 2018; Lynn, 1986). To determine the CVI, we employed the six-step process proposed by Yusoff (2019). This structured approach allowed for a thorough evaluation of each item’s relevance and clarity in relation to the concept being assessed.

5.1.4.1. Step I (preparation of content validation form for experts). The first step involved preparation of a structured content validation form to familiarize the experts with the task and expected outcomes. The form was provided along with detailed instructions, guiding experts on how to rate the relevance of each item to the construct using a four-point Likert scale (1 = not relevant to 4 = highly relevant), as suggested by Polit and Beck (2006). Additionally, a clear definition of the construct was shared to ensure consistency in expert understanding and ratings (Sultana & Farooq, 2025).

5.1.4.2. Step II (selection of experts). A panel of ten experts was purposively selected in accordance with established methodological recommendations (Polit et al., 2007). Expert selection followed predefined eligibility criteria to ensure both methodological rigor and substantive relevance. Specifically, panel members were required to (a) hold a doctoral degree, (b) demonstrate scholarly experience in scale development, psychometrics, or quantitative research methods, and/or (c) possess domain expertise in artificial intelligence in education, technology dependency, or related behavioral and psychological constructs. To preserve the integrity of the validation process, the expert panel was independent of the study participants, and no overlap between expert evaluators and respondent samples was permitted.

5.1.4.3. Step III (focus group discussion). A focus group discussion was conducted with the expert panel. Each expert received the validation form and was asked to assess the importance of each item with respect to the construct. This discussion was crucial for gathering preliminary feedback and ensuring that all dimensions of the construct were covered.

5.1.4.4. Step IV (scoring independently). Experts independently rated each item based on the domain definition in the validation form. This independent scoring minimized any potential biases and ensured an objective evaluation of each item.

5.1.4.5. Step V (form submission). After completing their assessments, experts submitted their forms to the researchers. This hands-off submission process ensured that the researchers only received final evaluations without influencing expert judgments.

5.1.4.6. Step VI (CVI computation): To evaluate content validity, we determined two indices.

- a. Item-level Content Validity Index (I-CVI): This index measures the content validity of individual items. It is calculated by dividing the number of experts who rated an item as relevant (3 or 4) by the total number of experts. An I-CVI score of 0.78 or higher is acceptable when more than nine experts are involved (Lynn, 1986).
- b. The Scale-level Content Validity Index (S-CVI): This index assesses the overall content validity of the scale. It is reported in two ways: S-CVI/Ave (average method) and S-CVI/UA (universal agreement method). In this evaluation, ratings of 3 or 4 were coded as 1, indicating relevance, while ratings of 1 or 2 were coded as 0, indicating non-relevance.

The level of agreement among experts was calculated by summing the number of experts who rated an item as relevant. For example, if all 8 experts deemed an item relevant, the total count of experts in agreement would be 10 (1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1). Conversely, universal agreement signifies unanimous consensus: if all experts agree on an item's relevance, the item receives a score of 1. However, if even one expert disagrees, the score drops to 0. This stringent criterion ensures that only items with broad consensus advance in the evaluation process.

5.1.5. Modified kappa (κ^*)

This statistic adjusts for the possibility that expert agreement among the panel members might occur by chance (Polit et al., 2007). It is calculated using the formula: $\kappa^* = \frac{ICVI - P_c}{1 - P_c}$ where P_c is the chance probability, computed as $P_c = (N!/A!(N-A)!)*0.5^N$ where N is the total number of experts and A is the number in agreement for a particular item. Only items with $I-CVI \geq 0.78$ and $\kappa^* > 0.75$ were considered for further validation.

5.1.6. S-CVI/UA

This index measures the proportion of items that achieved unanimous agreement among the experts. For instance, if 15 out of 25 items reached universal agreement among the 10 experts, the value would be 0.60. A lower S-CVI/UA score is expected when more experts are involved, as the likelihood of disagreement increases (Polit & Beck, 2006).

5.1.7. S-CVI/Ave based on I-CVI

This index averages the I-CVI scores across all items. A value of 0.923 (calculated by dividing the sum of all I-CVIs by the total number of items) exceeds the acceptable threshold of 0.90 (Waltz et al., 2010), confirming the scale's content validity. Ultimately, 25 items met the rigorous criteria ($I-CVI \geq 0.78$, $\kappa^* > 0.75$, and $S-CVI/Ave \geq 0.9$) and were selected for pilot testing. Table 2 presents the final retained items, along with their expert ratings and related calculations.

5.2. Phase 2: Scale purification

We conducted a thorough scale purification process and assessed the internal consistency of our measurement tool. During the scale development, we piloted a 25-item set using a self-administered questionnaire to collect primary data from our target respondents. The questionnaire was structured into two main sections: demographic variables and items specifically designed to measure AI dependency. Respondents provided feedback on a five-point Likert scale, ranging from 1 to 5. This format was chosen based on existing research, demonstrating its higher reliability, validity, and discriminatory power. The sample size consisted of 248 carefully selected respondents, ensuring it accurately reflected the larger population.

Before conducting exploratory factor analysis, we evaluated the suitability of our sample using the Kaiser-Meyer-Olkin (KMO) test and Bartlett's test of sphericity. The KMO value of 0.889 indicated adequate sampling, while Bartlett's test of sphericity was significant, confirming sufficient correlations within the data matrix. These results confirmed that our sample was appropriate for factor analysis.

We then proceeded to identify the underlying dimensions of the AI dependency scale through principal components analysis (PCA) with varimax rotation and Kaiser normalization, using SPSS v. 29.

Table 2. Content validity of the items selected to measure AI dependency.

Items	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	N	A	N-a	UA	I-cVI	Pc	K*	Interpretation
Item 1	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 2	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 3	1	1	1	1	1	0	1	0	1	1	10	8	2	0	0.8	0.043945	0.790807	Excellent
Item 4	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 5	1	1	1	1	1	1	1	0	1	1	10	9	1	0	0.9	0.009766	0.899014	Excellent
Item 6	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 7	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 8	1	1	1	1	1	1	1	1	1	0	10	9	1	0	0.9	0.009766	0.899014	Excellent
Item 9	1	1	1	1	1	1	1	1	0	1	10	9	1	0	0.9	0.009766	0.899014	Excellent
Item 10	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 11	1	1	1	1	0	1	1	1	1	1	10	9	1	0	0.9	0.009766	0.899014	Excellent
Item 12	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 13	1	1	1	1	1	0	1	1	1	1	10	9	1	0	0.9	0.009766	0.899014	Excellent
Item 14	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 15	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 16	0	1	1	1	1	1	1	1	1	1	10	9	1	0	0.9	0.009766	0.899014	Excellent
Item 17	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 18	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 19	1	1	1	1	1	0	1	1	0	1	10	8	2	0	0.8	0.043945	0.790807	Excellent
Item 20	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 21	1	1	1	1	1	1	1	1	0	1	10	9	1	0	0.9	0.009766	0.899014	Excellent
Item 22	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 23	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 24	1	1	1	1	1	1	1	1	1	1	10	10	0	1	1	0.000977	1	Excellent
Item 25	1	1	1	1	0	1	1	0	1	1	10	8	2	0	0.8	0.043945	0.790807	Excellent
															15	23.7		
Total Items																25		
S-CVI/AVE based on I-CVI																0.948		
S-CVI/UA																0.6		

Notes: N: Total no. of expert; A: No. of agreements; N-A: Total no. of expert (N) – No. of Agreements (A); UA: Universal agreement; Pc: Probability of chance agree; K*: Modified Kappa.

Table 3. Results of exploratory factor analysis for AI dependency scale.

Factor extracted	Item code	Items	Factor loading
Cognitive Reliance	COG1	I struggle to organize my research thoughts and ideas without AI assistance.	0.797
	COG2	I depend on AI to generate creative solutions to challenges I face in research.	0.762
	COG3	I often rely on AI to help me interpret complex research data.	0.800
	COG4	I often prefer using AI to assist in the report writing process rather than structuring it myself.	0.801
	COG5	I struggle to formulate research questions without input from AI tools.	0.797
Emotional Reliance	EMO1	I feel disappointed when I cannot find an AI tool for my task.	0.876
	EMO2	Become irritable if restricted from using AI tools.	0.736
	EMO3	I feel more confident in my research decisions when they are supported by AI data.	0.905
	EMO4	I get depressed when I realize that AI isn't contributing significantly to my project.	0.889
	EMO5	I would rather discuss my problems with AI (e.g., ChatGPT) than with my mentor or colleagues.	0.771
Behavioral reliance	BEHV1	I feel a strong urge to use AI even when it isn't necessary for the research task.	0.831
	BEHV2	Even when I decide to reduce my use of AI in research, I struggle to follow through.	0.867
	BEHV3	I have trouble stopping myself from using AI for research tasks I could do on my own.	0.760
	BEHV4	I rely on AI to verify research information, even when I already know the answers.	0.788
	BEHV5	I use AI tools daily for conducting research activities.	0.841

Items that did not meet best practices criteria were removed, resulting in a refined 15-item scale distributed across three discernible factors (Table 3). After five iterations, these factors accounted for 70.96% of the total variance, exceeding the benchmark threshold of 60%.

The reliability of the AI dependency scale was measured at 0.86, surpassing the recommended 0.7 cutoffs. Each individual dimension exhibited commendable internal consistency, with reliability coefficients ranging from 0.73 to 0.90. These emergent factors are named as cognitive reliance, emotional reliance, and behavioral Reliance based on their defining characteristics.

5.3. Phase 3: Scale refinement

5.3.1. Evaluation of latent factor structure

To enhance the dimensionality of the AI dependency scale, we conducted a confirmatory factor analysis to assess the model fit (Figure 2). This analysis was performed using SmartPLS 4.0 software, which

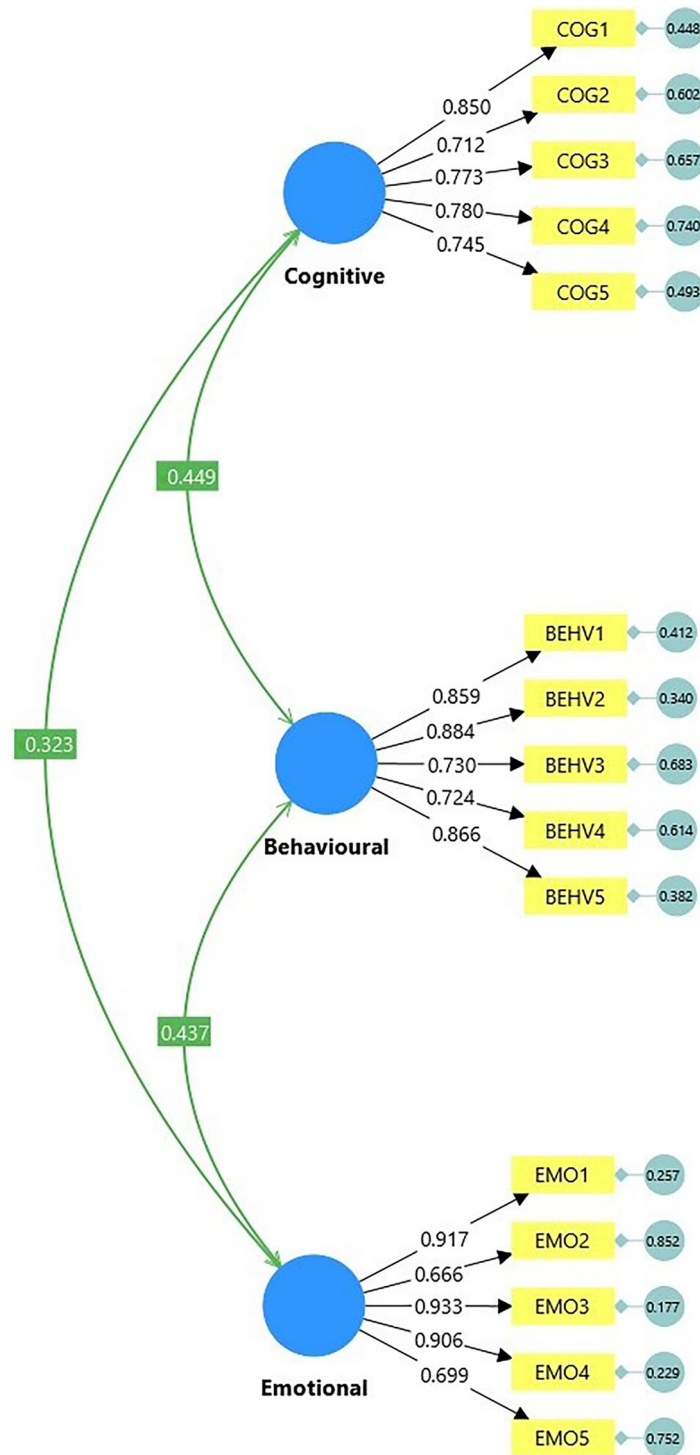


Figure 2. Measurement model for AI dependency scale using confirmatory factor analysis (CFA) (Authors findings).

facilitated the validation of the latent factor structure. A new dataset was collected from a sizeable sample of 219 academic scholars. A detailed profile of this sample is presented in [Table 1](#).

We employed hierarchical component modeling (HCM) to illustrate the relationship between the main concept and its indicators. In this setup, AI dependency is conceptualized as a Type 1 HCM, a reflective, second-order construct (Sarstedt et al., 2019), which encompasses Cognitive reliance, emotional reliance, and Behavioral Reliance as its foundational reflective constructs. To ensure precision in measuring our latent structure, we evaluated multicollinearity using the variance-inflation factor (VIF), with all VIF values were within the acceptable maximum benchmark of $VIF < 3.3$ (Diamantopoulos & Sigauw,

2006; Hair et al., 2019). All items demonstrated strong indicator loadings (ranging from 0.699 to 0.933) that exceeded the minimum standard of 0.6. Therefore, no item was considered for elimination. Both Cronbach's alpha (α) and composite reliability demonstrated values exceeding the minimum threshold of 0.7 (Bagozzi & Yi, 1988; Hinton et al., 2014; Nunally & Bernstein, 1978; Sarstedt et al., 2019). A detailed analysis of these results is presented in Table 4. The average variance extracted for the three dimensions was examined to determine convergent validity; all exceeded the minimum threshold of 0.5. Discriminant validity was confirmed using the Heterotrait-Monotrait (HTMT) criterion (Ahmad et al., 2025; Henseler et al., 2015), as shown in Table 5. The overall average variance extracted for AI dependency was 0.646, highlighting the coherence among its three dimensions. Additionally, the metrics from our confirmatory factor analysis indicated a satisfactory model fit, as evidenced by a root mean square residual (SRMR) value of 0.046.

Common method variance (CMV) is a significant concern in behavioral research, as measurement errors can introduce unnecessary variability not data. Common method variance (CMV) often arises when the same measurement method is used across multiple constructs, potentially leading to biased results (Podsakoff et al., 2003). To address this issue proactively, we implemented Harman's single factor test and the marker variable technique, as recommended by Lindell and Whitney (2001). Specifically, we employed Harman's single-factor matrix to evaluate CMV, following the Aulakh and Gencturk (2000) recommendation. The exploratory factor analysis (EFA) indicated that none of the four factors accounted for more than 50% of the variance, suggesting that our study did not exhibit common method bias, in line with the suggestion of Harman (1976).

Next, we identified "Brand Coolness" as a marker variable. This construct was measured using a six-item scale adapted from Loureiro et al. (2024) and assessed on a 5-point Likert scale ranging from 1- (strongly Agree) to 5 (Strongly disagree). The items effectively represented the variable and demonstrated high reliability ($\alpha = 0.826$). Importantly, the variance attributed to the marker variable remained low (below 25%), supporting our conclusion that CMV does not pose a threat to our study.

5.3.2. Tetrad analysis for model confirmation

To further confirm our model, we conducted a Confirmatory Tetrad Analysis (CTA-PLS) as outlined by Hair et al. (2019). This analysis examines tetrad inconsistencies among indicator covariance pairs, providing insights into whether constructs are formative or reflective. Our results revealed that the confidence intervals of these tetrads included zero, indicating statistical insignificance (Bollen & Ting, 2000; Noor et al., 2022). This lends empirical support to the notion that reliance on AI should be treated a reflective construct.

5.4. Phase 4: Association between AI dependency and research scholars' attitudes toward plagiarism

This section addresses the second research question (RQ2) by examining the association between AI dependency and research scholars' attitudes toward plagiarism. In line with this objective, the nomological validity of the AI Dependency Scale was assessed by situating the construct within a theoretically grounded network of related variables and evaluating whether it demonstrates theoretically consistent

Table 4. Construct reliability and validity of AI dependency scale.

Sub-dimension	Cronbach's alpha	Composite reliability (CR)	Average variance extracted (AVE)
Cognitive reliance	0.892	0.894	0.581
Emotional reliance	0.915	0.915	0.693
Behavioral reliance	0.907	0.910	0.665

Table 5. Hetetrotrait-Monotrait (HTMT) ratio of correlations AI dependency dimensions.

Dimensions	Behavioral reliance	Cognitive reliance	Emotional reliance
Cognitive reliance	0.441	1	0.348
Emotional reliance	0.470	0.348	1
Behavioral reliance	1	0.441	0.470

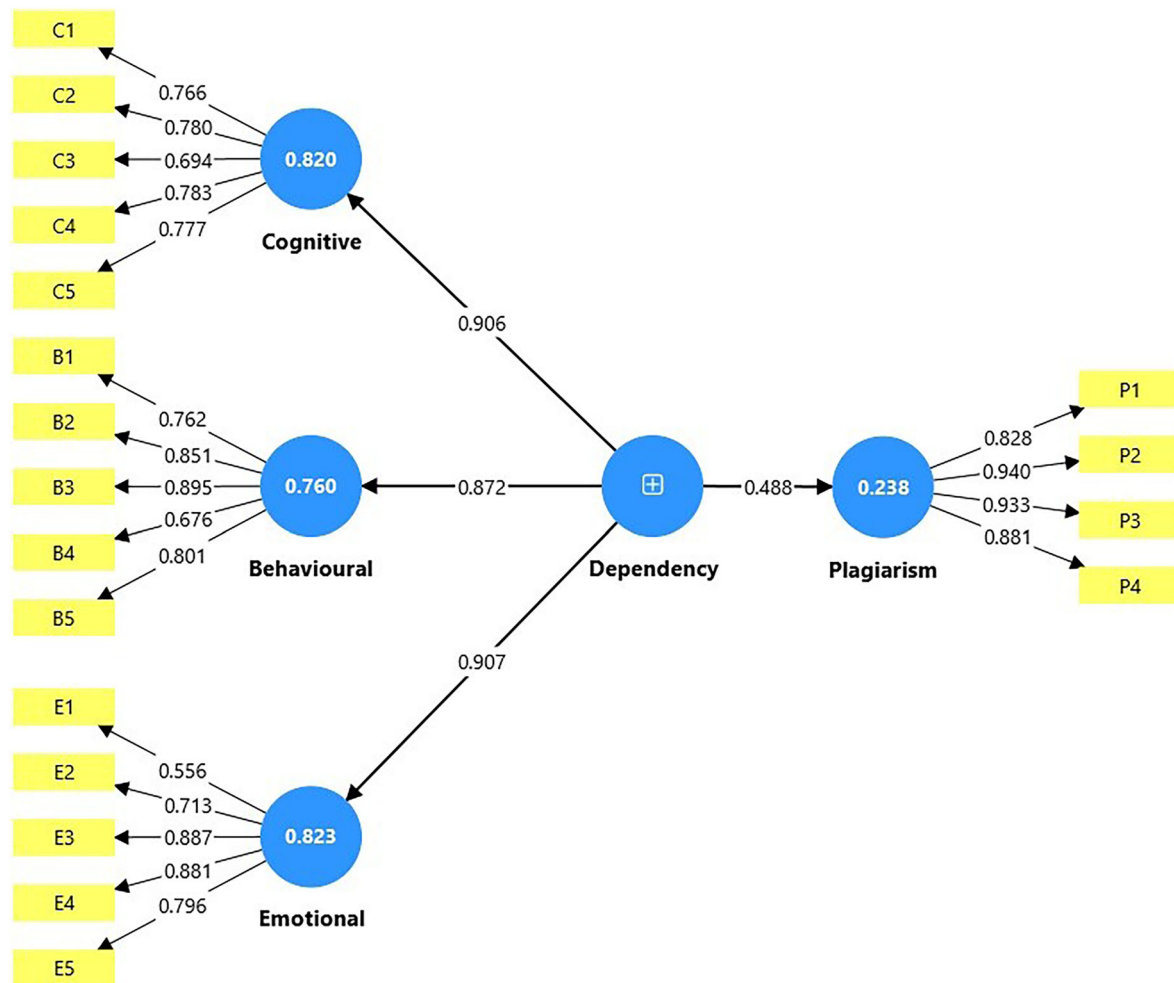


Figure 3. Structural equation model to assess the predictive strength of AI dependency with plagiarism (author's finding).

associations within its domain. Nomological validity refers to the extent to which a construct behaves in accordance with established theoretical expectations by exhibiting meaningful relationships with conceptually related constructs. Accordingly, attitudes toward plagiarism were incorporated as a theoretically relevant construct within the nomological network of AI dependency, consistent with prior work highlighting ethical orientations as an important correlate of technology reliance (Yeo, 2023). The plagiarism attitude items were adapted from Farooq and Sultana (2022) and measured using a five-point Likert scale. This phase of analysis was conducted using a new dataset comprising 200 participants whose demographic characteristics closely matched those of the samples employed in the exploratory and confirmatory factor analyses, thereby supporting consistency across validation stages.

The ethical implications of AI use in educational contexts have received increasing scholarly attention. As AI tools become more deeply embedded in learning and research environments, reliance on these technologies has emerged as a salient factor in discussions of academic integrity. Theoretical and empirical research suggests that excessive dependence on AI may be associated with more permissive attitudes toward plagiarism, particularly due to the ease of information access and the potential normalization of ethically ambiguous practices (Dawson, 2023; Elali & Rachid, 2023). Conversely, educational systems that promote balanced and responsible AI use not only enhance learning efficiency but also contribute to the preservation of academic integrity (Rodrigues et al., 2025). Individuals who recognize the limitations of AI assistance and engage with such tools reflectively are more likely to develop a stronger ethical orientation, thereby reducing susceptibility to academic misconduct (Feng, Wang, et al., 2024). Prior studies

Table 6. Hypothesis testing of nomological validity of AI dependency scale with plagiarism.

Hypothesis	Relationship	Original sample	Sample mean	Standard deviation	t-Statistic	p-Value	Effect	Outcome
H1	Dep → Plag	0.488	0.486	0.010	48.800	0.000	Significant	Supported

further underscore that responsible technology use and a clear understanding of academic integrity play a crucial role in shaping attitudes toward plagiarism (Feng, Zhou, et al., 2024; Perkins & Roe, 2024; Striepe et al., 2023). Based on these theoretical considerations, the following hypothesis was formulated: AI dependency is positively associated with attitudes toward plagiarism.

To empirically examine this hypothesis, structural equation modeling was employed to assess the association between AI dependency and attitudes toward plagiarism. The analysis revealed a statistically significant relationship between the two constructs ($\beta = 0.488$, $p < 0.001$), as illustrated in Figure 3 and summarized in Table 6. This finding provides empirical support for the proposed hypothesis and demonstrates that AI dependency exhibits theoretically consistent associations with plagiarism attitudes, thereby supporting the nomological validity of the AI Dependency Scale. The observed relationship aligns with prior literature at the intersection of technology use and academic ethics, which highlights how engagement with digital tools can shape ethical orientations in educational and research contexts (Brown et al., 2018; Johnson, 2023). Collectively, these results situate AI dependency within its expected nomological network and strengthen confidence in the theoretical coherence of the scale.

5.5. Phase 5: Cross-sample validation and generalizability of the scale

A crucial indicator of a scale's robustness lies in its ability to reproduce consistent psychometric properties across independent samples. In this study, generalizability refers to the cross-sample stability of the AI Dependency Scale, specifically, its capacity to replicate the same factor structure, reliability, and validity indices in a separate, larger cohort of research scholars drawn from multiple disciplines. This approach aligns with standard practices in early-stage scale development, where demonstrating reproducibility across samples strengthens confidence in the instrument's measurement precision. Cross-sample validation is a critical indicator of scale robustness, as it demonstrates that the instrument performs consistently beyond the initial development samples and across heterogeneous populations.

To assess this, we collected data from an additional independent sample of 588 research scholars enrolled across public, private, and deemed universities. This sample represented diverse fields, including sciences, social sciences, humanities, engineering, and medical studies, mirroring the disciplinary breadth of earlier phases while substantially expanding the participant base.

Participant recruitment in Phase 5 followed the same on-site, paper-based procedure used in Phases 2–4 (Section 4.2). Potential respondents were approached in doctoral research spaces, provided with a brief explanation of the study, and invited to participate voluntarily after giving informed consent. Eligibility criteria remained identical, restricting participation to actively enrolled PhD scholars engaged in academic research. To ensure that respondents from earlier phases were not included in this replication sample, several safeguards were implemented. First, Phase 5 data collection was conducted during a separate recruitment wave and intentionally prioritized departments and institutions that were not used in the earlier phases. Although the same categories of universities (public, private, and deemed) were included to maintain institutional diversity, Phase 5 recruitment deliberately involved different universities and academic departments from those sampled previously. Second, participant eligibility for Phase 5 was verified through a structured self-report screening procedure administered at the point of data collection. As part of the informed consent and study briefing process, participants were explicitly asked whether they had participated in any earlier phase of the study. Individuals who reported prior participation were excluded from the replication sample and did not proceed to survey completion. The use of structured self-report screening as an exclusion criterion is a standard practice in multi-phase survey research and scale validation studies, particularly when data are collected across independent recruitment waves and institutional sites (Dillman et al., 2014; Fowler, 2013). When combined with separate recruitment waves and sampling from different institutions, this approach is widely considered

Table 7. Construct reliability and validity of AI dependency scale.

Sub-dimension	Cronbach's alpha	Composite reliability (CR)	Average variance extracted (AVE)
Cognitive Reliance	0.899	0.903	0.637
Emotional Reliance	0.920	0.923	0.706
Behavioral Reliance	0.896	0.897	0.641

appropriate for minimizing sample overlap and ensuring the independence of validation cohorts (DeVellis, 2016; Worthington & Whittaker, 2006).

Confirmatory factor analysis demonstrated an excellent model fit ($SRMR = 0.044$), and all reliability indicators, including Cronbach's α , composite reliability, and standardized loadings, met recommended thresholds (Table 7). Convergent validity was reaffirmed through AVE values exceeding 0.50, and discriminant validity was confirmed with HTMT ratios below 0.85 (Hair et al., 2019). These indicators collectively confirm that the proposed multidimensional structure is stable and psychometrically sound across independent samples. As it demonstrated satisfactory model fit and stable factor loadings, supporting the scale's capacity to measure AI dependency consistently across varied educational and disciplinary settings.

Through this replication in a larger and heterogeneous cohort, the 15-item AI Dependency Scale demonstrates strong cross-sample generalizability, reinforcing its suitability for measuring AI dependency among research scholars across varied disciplinary backgrounds in the higher education context.

6. Discussion

We set out to develop and validate a comprehensive AI Dependency Scale designed to capture the different dimensions of AI reliance in academic settings. Through exploratory factor analysis, we identified three distinct factors representing cognitive, emotional, and habitual behaviors associated with AI use. Confirmatory factor analysis further validated the scale's robustness and reliability. These results show that the scale accurately captures the multi-dimensional nature of AI dependency and is applicable across diverse academic research contexts.

The dimension of *cognitive reliance* assesses the extent to which researchers rely on AI to perform intellectual tasks, such as data analysis, problem-solving, and decision-making. The widespread use of AI for cognitive support signals a shift in the academic ecosystem, positioning AI tools as central to the intellectual processes of researchers. For instance, items such as "I often rely on AI to help me interpret complex research data" illustrate how deeply integrated AI has become in scholars' cognitive workflows. This integration fosters greater efficiency in managing large datasets and tackling complex problems. However, this dependency also carries risks. An over-reliance on AI for cognitive tasks may erode critical thinking and independent decision-making, potentially weakening researchers' analytical and reasoning skills over time.

The second dimension, *emotional reliance*, explores the emotional bonds that researchers form with AI tools. Emotions such as frustration when AI is unavailable or confidence when AI supports decision-making demonstrate that AI tools serve not only functional purposes but also have a significant impact on researchers' emotional well-being. For example, the statement "I feel more confident in my research decisions when they are supported by AI data" shows how AI tools can bolster researchers' confidence in their work. However, emotional reliance can have unintended consequences. When researchers develop an emotional dependency on AI, they may experience a diminished sense of self-efficacy regarding their ability to produce independent work without AI support. This could result in issues such as academic dishonesty or a compromise in research integrity.

The *habitual behavioral reliance* captures the degree to which AI use becomes an automatic and recurring part of researchers' everyday academic routines. Items such as "I use AI tools daily for my research activities," "I feel a strong urge to use AI even when it isn't necessary," and "I struggle to reduce my use of AI" collectively illustrate how AI shifts from a deliberate choice to a habitual behavioral pattern. Although such routines can streamline workflows and improve efficiency, they also raise questions about over-automation and the gradual erosion of independent academic effort. When researchers default to AI even for tasks they could perform unaided the behavior suggests a

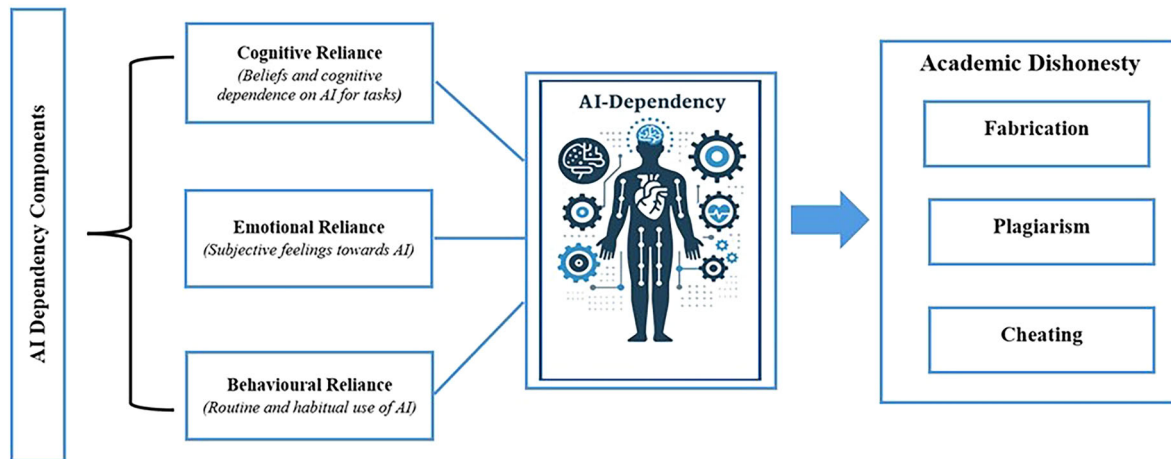


Figure 4. Components of AI-dependency and its influence on academic integrity (proposed model by authors).

deeply ingrained reliance that may limit opportunities for creative thinking, problem-solving, and critical analysis. Thus, habitual reliance reflects not only frequent use but the potential risk of researchers allowing automated assistance to replace essential scholarly skills over time.

The dimensions of AI dependency, including cognitive reliance, emotional reliance, and behavioral reliance, can collectively contribute to an over-reliance on AI, which may ultimately lead to academic dishonesty. Cognitive reliance, where researchers heavily rely on AI for intellectual tasks like data analysis and decision-making, can undermine critical thinking and independent reasoning skills over time. As researchers shift toward automated cognitive processes, they may find their analytical abilities eroded, making them more vulnerable to unethical practices, such as plagiarism or fabrication.

Similarly, emotional reliance reflects the strong connections researchers develop with AI tools, which can boost their confidence when supported by AI-generated insights. However, this emotional bond can result in a diminished belief in their ability to complete work independently. As researchers increasingly view AI as the primary driver of their work rather than a collaborative partner, they may engage in dishonest behaviors to meet academic expectations (Salah et al., 2024). Lastly, habitual behavioral reliance encompasses the ingrained patterns of AI usage in daily academic activities. The frequent use of AI for routine tasks may create a preference for automation over manual intellectual labor. This reliance may stifle creativity and critical thinking, pushing researchers toward unethical practices, such as cheating, plagiarism, and fabrication, to meet academic demands.

The growing dependence on AI tools increases the likelihood of academic dishonesty (Cotton et al. 2024). Figure 4 illustrates the potential consequences of AI dependency, pointing to specific unethical behaviors such as plagiarism (Yeo, 2023), fabrication (Currie, 2023), and cheating (Cotton et al., 2024). Future research could expand this model to explore additional unethical behaviors and provide a more comprehensive understanding of how AI influences academic integrity. These findings emphasize the importance of developing guidelines that promote responsible AI use to safeguard academic integrity in increasingly AI-driven academic environments.

7. Practical implications

The findings of this study carry important practical implications for educational institutions, policy-makers, and academic communities navigating the growing integration of generative AI tools.

7.1. Institutional strategies for ethical AI integration

Educational institutions should develop comprehensive AI literacy programs and establish clear guidelines for ethical AI use. By setting standards for responsible AI engagement and promoting transparency in its application, institutions can mitigate risks such as academic dishonesty and foster a culture of ethical innovation.

7.2. Promoting balanced AI use and academic integrity

While AI tools enhance productivity, unchecked reliance can undermine academic rigor. Embedding critical AI literacy in curricula alongside activities that promote independent thinking can help students and researchers maintain academic standards while benefiting from AI support.

7.3. Supporting emotional and cognitive well-being

Institutions should recognize the emotional and psychological impacts of AI dependency, including over-reliance and anxiety. Offering workshops, counseling, and peer mentoring programs can help academic users navigate their relationship with AI tools more healthily.

7.4. Encouraging cognitive autonomy and innovation

To prevent cognitive complacency, educators should design assignments that require comparative analysis of AI-generated versus human-produced outputs. This not only reinforces research integrity but also stimulates critical evaluation skills.

7.5. Fostering a culture of lifelong AI literacy

As AI continues to evolve, institutions must promote continuous learning and adaptation. Regular training sessions, seminars, and institutional dialogues on the responsible use of AI will ensure that academic communities remain both ethically and technologically informed.

Together, these implications emphasize the need for proactive and ethically grounded approaches to managing AI integration in academia, ensuring that GenAI tools enhance rather than hinder academic development and scholarly authenticity.

8. Limitations and directions for future research

This study presents a validated and theoretically grounded instrument for assessing AI dependency; several areas remain open for further investigation and refinement. First, the scale was validated with academic researchers, which was appropriate given the study's focus on scholarly engagement with generative AI. However, the generalizability of findings to other academic populations, such as undergraduate and postgraduate students, remains limited. Future research should consider testing the scale across a broader range of learner groups to explore potential differences in dependency patterns based on academic level or role.

Second, while the scale captures psychological and behavioral dimensions of dependency, it does not incorporate dispositional traits such as impulsivity or neuroticism, which are emphasized in the I-PACE model. Future studies could explore the role of personality traits in shaping AI dependency, particularly within predictive frameworks or structural equation modeling. Third, the current study focuses on plagiarism as a key outcome of AI overreliance. However, other forms of academic misconduct, such as unauthorized collaboration, ghost authorship, or content fabrication, may also be influenced by the use of AI. Expanding the outcome space could enrich our understanding of the broader ethical risks associated with generative AI in academic contexts.

Finally, the study was conducted within the Indian higher-education context, which may limit the immediate generalizability of the findings. However, because the scale is grounded in well-established psychological theories that are not culturally specific and because research scholars across countries operate within broadly similar academic environments shaped by peer review, publication expectations, intellectual labor, and increasing exposure to AI tools the underlying structure of AI dependency is likely applicable across contexts. Future research should therefore adapt and validate the scale in other countries and institutional settings. Such cross-cultural and multi-institutional validation using diverse international samples would further strengthen the scale's robustness, applicability, and empirical reach.

9. Conclusion

This study introduces a validated instrument to measure AI dependency in academic research environments, conceptualized through cognitive, emotional, and behavioral dimensions. By integrating theoretical frameworks from psychological and educational research, the AI Dependency Scale advances current understanding of how scholars engage with generative AI tools. The findings not only demonstrate the scale's psychometric robustness but also highlight its relevance in predicting ethical vulnerabilities, particularly permissive attitudes toward plagiarism. In doing so, this research provides a critical resource for future inquiry into technology-mediated academic behavior and offers a foundation for institutional policies that promote ethical and balanced AI usage. The scale stands to support educators, researchers, and administrators in navigating the complex intersection of academic productivity, psychological reliance, and digital responsibility in the age of artificial intelligence.

This study addresses the growing concern about academic researchers' psychological reliance on generative AI tools by developing and validating a multidimensional AI Dependency Scale. Drawing on the I-PACE model, Basic Psychological Needs Theory, and the ABC model of attitude structure, the scale captures three core dimensions of dependency: cognitive, emotional, and behavioral reliance. Unlike previous unidimensional instruments, this scale addresses a critical gap by offering a theoretically grounded and empirically validated tool tailored to the context of higher education research.

The findings reinforce that dependency on AI is not merely a function of usage frequency but is deeply intertwined with affective states, cognitive habits, and academic pressures. Importantly, the study establishes a significant link between AI dependency and self-reported permissiveness toward academic dishonesty, highlighting the ethical implications of unchecked reliance. By providing a robust framework for measuring AI dependency, this research advances the discourse on digital ethics, academic integrity, and the responsible integration of AI. It offers not only a diagnostic tool for educational institutions but also a foundation for designing interventions that foster critical engagement and ethical judgment in AI-mediated scholarship. As the academic landscape continues to evolve, this scale enables researchers and educators to monitor, understand, and guide the psychological dynamics that shape the future of research practice.

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Ethical approval

The study received ethical approval from the Institutional Review Board of Jain University, Bengaluru, India (IRB No: IRB-25-11-407). This study involved human participants and adhered to all relevant ethical guidelines. Informed consent was obtained from all participants prior to data collection.

Consent for publication

All authors have read and approved the final version of the manuscript and consent to its publication in the International Journal of Educational Integrity.

Author contributions

CRediT: **Almaas Sultana**: Conceptualization, Formal analysis, Methodology, Software, Validation, Writing – original draft, Writing – review & editing; **Mugaahed Abdu Kaid Saleh**: Data curation, Investigation; **Abdellatif Sellami**: Writing – review & editing; **Zubair Ahmad**: Methodology, Validation, Writing – review & editing.

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References

- Abbas, S. G., Ehsan, M., Akbar, G., Rehman, A., Bibi, H., & Sian, Z. H. (2023). Effects of ChatGPT integration as an artificial intelligence tool for education and research: An exploratory survey at the university level. *PalArch's Journal of Archaeology of Egypt/Egyptology*, 20(2), 1993–2017.
- Ahmad, Z., Sultana, A., Siby, N., Ammar, M., & Al-Thani, N. J. (2025). A multi-dimensional scale for measuring undergraduates' interest in STEM disciplines. *Journal of Science Education and Technology*, 34(4), 611–620. <https://doi.org/10.1007/s10956-024-10140-w>
- Ali, K., Garcia, A., & Vadsariya, A. (2024). Impact of the AI dependency revolution on both physical and mental health. *Journal of Strategic Innovation and Sustainability*, 19(2), 70–81. <https://doi.org/10.33423/jsis.v19i2.7006>
- Al-Zahrani, A. M. (2024). The impact of generative AI tools on researchers and research: Implications for academia in higher education. *Innovations in Education and Teaching International*, 61(5), 1029–1043. <https://doi.org/10.1080/14703297.2023.2271445>
- Aulakh, P. S., & Gencturk, E. F. (2000). International principal–agent relationships: Control, governance and performance. *Industrial Marketing Management*, 29(6), 521–538. [https://doi.org/10.1016/S0019-8501\(00\)00126-7](https://doi.org/10.1016/S0019-8501(00)00126-7)
- Bacallo, D. L. D. L., Bie, P. E. G., Tulabut, R. Y. C., Pineda, A. J. M., Vicente, A. P., De Vera, A. R., Sawal, E. J. F., Marcaida, J. L. M., & Dizon, R. C. (2024). Digital learning revolution: Identifying the influence of dependency on artificial intelligence tools towards the knowledge acquisition of students. *Ignatian International Journal for Multidisciplinary Research*, 2(8), 1338–1362. <https://doi.org/10.5281/zenodo.13359360>
- Bagozzi, R. P., & Yi, Y. (1988). On the evaluation of structural equation models. *Journal of the Academy of Marketing Science*, 16(1), 74–94. <https://doi.org/10.1007/BF02723327>
- Ball-Rokeach, S. J., & DeFleur, M. L. (1976). A dependency model of mass-media effects. *Communication Research*, 3(1), 3–21. <https://doi.org/10.1177/009365027600300>
- Bányai, F., Zsila, Á., Király, O., Maraz, A., Elekes, Z., Griffiths, M. D., Andreassen, C. S., & Demetrovics, Z. (2017). Problematic social media use: Results from a large-scale nationally representative adolescent sample. *PloS One*, 12(1), e0169839. <https://doi.org/10.1371/journal.pone.0169839>
- Boateng, G. O., Martin, S. L., Collins, S. M., Natamba, B. K., & Young, S. L. (2018). Measuring exclusive breastfeeding social support: Scale development and validation in Uganda. *Maternal & Child Nutrition*, 14(3), e12579. <https://doi.org/10.1111/mcn.12579>
- Bollen, K. A., & Ting, K. F. (2000). A tetrad test for causal indicators. *Psychological Methods*, 5(1), 3–22. <https://doi.org/10.1037/1082-989X.5.1.3>
- Brand, M., Müller, A., Wegmann, E., Antons, S., Brandtner, A., Müller, S. M., Stark, R., Steins-Loeber, S., & Potenza, M. N. (2025). Current interpretations of the I-PACE model of behavioral addictions. *Journal of Behavioral Addictions*, 14(1), 1–17. <https://doi.org/10.1556/2006.2025.00020>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- Brown, T., Bourke-Taylor, H., Isbel, S., Gustafsson, L., McKinstry, C., Logan, A., & Etherington, J. (2018). Exploring similarities and differences among the self-reported academic integrity of Australian occupational therapy domestic and international students. *Nurse Education Today*, 70(11), 13–19. <https://doi.org/10.1016/j.nedt.2018.08.005>
- Capinding, A. T. (2024). Students' AI dependency in 3R's: Questionnaire construction and validation. *International Journal of Information and Education Technology*, 14(11), 1532–1543. <https://doi.org/10.18178/ijiet.2024.14.11.2184>
- Carpenter, S. J. (2018). Ten steps in scale development and reporting: A guide for researchers. *Communication Methods and Measures*, 12(1), 25–44. <https://doi.org/10.1080/19312458.2017.1396583>
- Churchill, G. A. Jr. (1979). A paradigm for developing better measures of marketing constructs. *Journal of Marketing Research*, 16(1), 64–73. <https://doi.org/10.1177/002224377901600110>
- Clemmensen, T. H., Kristensen, H. K., Andersen-Ranberg, K., & Lauridsen, H. H. (2021). Development and field-testing of the Dementia Carer Assessment of Support Needs Tool (DECANT). *International Psychogeriatrics*, 33(4), 405–417. <https://doi.org/10.1017/S1041610220000114>
- Colella, S. (2025). The language of the digital air: AI-generated literature and the performance of authorship. *Humanities*, 14(8), 164. <https://doi.org/10.3390/h14080164>

- Cotton, D. R., Cotton, P. A., & Shipway, J. R. (2024). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education and Teaching International*, 61(2), 228–239. <https://doi.org/10.1080/14703297.2023.2184008>
- Currie, G. M. (2023). Academic integrity and artificial intelligence: Is ChatGPT hype, hero or heresy? *Seminars in Nuclear Medicine*, 53(5), 719–730. <https://doi.org/10.1053/j.semnuclmed.2023.04.003>
- Dawson, A. G. (2023). *Artificial intelligence and academic integrity*. Aspen Publishing.
- de Winter, J., Hancock, P. A., & Eisma, Y. B. (2025). ChatGPT and academic work: New psychological phenomena. *AI & Society*, 40(6), 4855–4868. <https://doi.org/10.1007/s00146-025-02241-w>
- Deci, E. L., & Ryan, R. M. (2000). The “what” and “why” of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry*, 11(4), 227–268. https://doi.org/10.1207/S15327965PLI1104_01
- Désiron, J. C., & Petko, D. (2023). Academic dishonesty when doing homework: How digital technologies are put to bad use in secondary schools. *Education and Information Technologies*, 28(2), 1251–1271. <https://doi.org/10.1007/s10639-022-11231-0>
- DeVellis, R. F. (2016). *Scale development: Theory and applications* (vol. 26). Sage publications.
- Diamantopoulos, A., & Siguaw, J. A. (2006). Formative versus reflective indicators in organizational measure development: A comparison and empirical illustration. *British Journal of Management*, 17(4), 263–282. <https://doi.org/10.1111/j.1467-8551.2006.00500.x>
- Dillman, D. A., Smyth, J. D., & Christian, L. M. (2014). *Internet, phone, mail, and mixed-mode surveys: The tailored design method* (4th ed.). John Wiley & Sons.
- Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., ... Wright, R. (2023). Opinion Paper: “So what if ChatGPT wrote it?” Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, 71(4), 102642. <https://doi.org/10.1016/j.ijinfomgt.2023.102642>
- Eagly, A. H., & Chaiken, S. (1993). *The psychology of attitudes*. Harcourt Brace Jovanovich College Publishers.
- Elali, F. R., & Rachid, L. N. (2023). AI-generated research paper fabrication and plagiarism in the scientific community. *Patterns*, 4(3), 100706. <https://doi.org/10.1016/j.patter.2023.100778>
- Else, H. (2023). By Chatgpt fool scientists. *Nature*, 613(7944), 423–423. <https://doi.org/10.1038/d41586-023-00056-7>
- Farooq, R., & Sultana, A. (2022). Measuring students’ attitudes toward plagiarism. *Ethics & Behavior*, 32(3), 210–224. <https://doi.org/10.1080/10508422.2021.1893752>
- Feng, W., Zhou, L., & Xiao, J. (2024). Beyond the traditional: Extending academic libraries’ roles in research integrity based on the causes of research misconduct. *The Journal of Academic Librarianship*, 50(3), 102877. <https://doi.org/10.1016/j.acalib.2024.102877>
- Feng, Y., Wang, J., & Anderson, S. G. (2024). Ethical considerations in integrating AI in research consultations: Assessing the possibilities and limits of GPT-based chatbots. *Journal of eScience Librarianship*, 13(1), e2024012. <https://doi.org/10.7191/jeslib.2024.2024012>
- Fowler, F. J. Jr. (2013). *Survey research methods*. Sage publications.
- Fui-Hoon Nah, F., Siau, K., & Shen, Z. (2023). Generative artificial intelligence (GenAI) in education: A framework and implications for research and practice. *Journal of Educational Technology & Society*, 26(3), 1–16. [https://doi.org/10.30191/ETS.202307_26\(3\).0001](https://doi.org/10.30191/ETS.202307_26(3).0001)
- Gillath, O., Ai, T., Branicky, M. S., Keshmiri, S., Davison, R. B., & Spaulding, R. (2021). Attachment and trust in artificial intelligence. *Computers in Human Behavior*, 115(2), 106607. <https://doi.org/10.1016/j.chb.2020.106607>
- Hair, J. F., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2019). When to use and how to report the results of PLS-SEM. *European Business Review*, 31(1), 2–24. <https://doi.org/10.1108/EBR-11-2018-0203>
- Hanafi, A. M., Ahmed, M. s., Al-Mansi, M. M., & Al-Sharif, O. A. (2025). Generative AI in academia: A comprehensive review of applications and implications for the research process. *International Journal of Engineering and Applied Sciences-October 6 University*, 2(1), 91–110. <https://doi.org/10.21608/ijeasou.2025.349520.1041>
- Harman, H. H. (1976). *Modern factor analysis*. (3rd ed.). University of Chicago Press.
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135. <https://doi.org/10.1007/s11747-014-0403-8>
- Hinkin, T. R. (1998). A brief tutorial on the development of measures for use in survey questionnaires. *Organizational Research Methods*, 1(1), 104–121. <https://doi.org/10.1177/109442819800100>
- Hinton, P. R., McMurray, I., & Brownlow, C. (2014). *SPSS explained*. Routledge.
- Igbokwe, I. C. (2023). Application of artificial intelligence (AI) in educational management. *International Journal of Scientific and Research Publications*, 13(3), 300–307. <https://doi.org/10.29322/IJSRP.13.03.2023.p13538>
- Jacovi, A., Marasović, A., Miller, T., & Goldberg, Y. (2021, March). Formalizing trust in artificial intelligence: Prerequisites, causes and goals of human trust in AI. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FACt ’21)* (pp. 624–635). Association for Computing Machinery. <https://doi.org/10.1145/3442188.3445923>

- Ji, Y., Zhong, M., Lyu, S., Li, T., Niu, S., & Zhan, Z. (2025). How does AI literacy affect individual innovative behavior: The mediating role of psychological need satisfaction, creative self-efficacy, and self-regulated learning. *Education and Information Technologies*, 30(11), 16133–16162. <https://doi.org/10.1007/s10639-025-13437-4>
- Johnson, C. (2023). *Understanding academic integrity and plagiarism in the digital age: Can digital forensics techniques help prevent and detect academic misconduct?* University of South Wales.
- Jorolan, J., Cabillo, F., Batucan, R. R., Camansi, C. M., Etoquilla, A., Gapo, J., Hallarte, D. K., Milano, M. L., Gonzales, R., & Gonzales, G. (2025). Development and validation of moral ascendancy and dependency in AI integration scale (MAD-AI) for teachers. *Computers & Education*, 235(12), 105346. <https://doi.org/10.1016/j.compedu.2025.105346>
- Khalaf, M. A. (2025). Does attitude towards plagiarism predict aigiarism using ChatGPT? *AI and Ethics*, 5(1), 677–688. <https://doi.org/10.1007/s43681-024-00426-5>
- King, M. R., ChatGPT. (2023). A conversation on artificial intelligence, chatbots, and plagiarism in higher education. *Cellular and Molecular Bioengineering*, 16(1), 1–2. <https://doi.org/10.1007/s12195-023-00730-4>
- Klingbeil, A., Grützner, C., & Schreck, P. (2024). Trust and reliance on AI-An experimental study on the extent and costs of overreliance on AI. *Computers in Human Behavior*, 160(11), 108352. <https://doi.org/10.1016/j.chb.2024.108352>
- Kotsis, K. T. (2024). Artificial intelligence creates plagiarism or academic research? *European Journal of Arts, Humanities and Social Sciences*, 1(6), 169–179. [https://doi.org/10.59324/ejahss.2024.1\(6\).18](https://doi.org/10.59324/ejahss.2024.1(6).18)
- Kovacic, D. (2018). Using the content validity index to determine content validity of an instrument assessing health care providers' general knowledge of human trafficking. *Journal of Human Trafficking*, 4(4), 327–335. <https://doi.org/10.1080/23322705.2017.1364905>
- Kwon, M., Lee, J.-Y., Won, W.-Y., Park, J.-W., Min, J.-A., Hahn, C., Gu, X., Choi, J.-H., & Kim, D.-J. (2013). Development and validation of a smartphone addiction scale (SAS). *PLOS One*, 8(2), e56936. <https://doi.org/10.1371/journal.pone.0056936>
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174. <https://doi.org/10.2307/2529310>
- Lindell, M. K., & Whitney, D. J. (2001). Accounting for common method variance in cross-sectional research designs. *The Journal of Applied Psychology*, 86(1), 114–121. <https://doi.org/10.1037/0021-9010.86.1.114>
- Lo, D. (2023, May). Trustworthy and synergistic artificial intelligence for software engineering: Vision and roadmaps. In *Proceedings of the IEEE/ACM International Conference on Software Engineering: Future of Software Engineering (ICSE-FoSE 2023)* (pp. 69–85). IEEE. <https://doi.org/10.1109/ICSE-FoSE57868.2023.00015>
- Loureiro, S. M. C., Aleem, A., & Breazeale, M. (2024). Luxury fashion brand coolness: Niche versus mass cool. *Spanish Journal of Marketing - ESIC*, 28(2), 165–186. <https://doi.org/10.1108/SJME-11-2023-0225>
- Lund, B. D., Wang, T., & Mannuru, N. R. (2023). Generative artificial intelligence in higher education: A qualitative exploration of instructors' perspectives. *Education and Information Technologies*, 28(11), 13383–13403. <https://doi.org/10.1007/s10639-023-12169-w>
- Lynn, M. R. (1986). Determination and quantification of content validity. *Nursing Research*, 35(6), 382–385. <https://doi.org/10.1097/00006199-198611000-00017>
- Mishra, M., & Anthony, M. (2024). Behavioural intention and dependency on ChatGPT among college students: Scale development and validation. *International Journal of Science and Research*, 13(4), 1098–1101. <https://doi.org/10.21275/SR24411155305>
- Morales-García, W. C., Sairitupa-Sanchez, L. Z., Morales-García, S. B., & Morales-García, M. (2024). Development and validation of a scale for dependence on artificial intelligence in university students. *Frontiers in Education*, 9, 1323898. <https://doi.org/10.3389/feduc.2024.1323898>
- Naseer, A., Ahmad, N. R., & Chishti, M. A. (2025). Psychological impacts of AI dependence: Assessing the cognitive and emotional costs of intelligent systems in daily life. *Review of Applied Management and Social Sciences*, 8(1), 291–307. <https://doi.org/10.47067/ramss.v8i1.458>
- Netemeyer, R. G., Bearden, W. O., & Sharma, S. (2003). *Scaling procedures: Issues and applications*. Sage Publications.
- Noor, N., Hill, S. R., & Troshani, I. (2022). Developing a service quality scale for artificial intelligence service agents. *European Journal of Marketing*, 56(5), 1301–1336. <https://doi.org/10.1108/EJM-03-2020-0179>
- Nunally, J. C., & Bernstein, I. H. (1978). *Psychometric theory*. McGraw-Hill.
- O'Connor, A. J. (2024). *Organizing for generative AI and the productivity revolution: Reshaping organizational roles in the age of artificial intelligence*. Springer Nature.
- Ogwueleka, F. N. (2025). Plagiarism detection in the age of artificial intelligence: Current technologies and future directions. In P. A. Okebukola (Ed.), *Handbook on artificial intelligence and quality higher education: AI and ethics, academic integrity and the future of quality assurance in higher education* (Vol. 3, pp. 120–129).
- Papadas, K. K., Avlonitis, G. J., & Carrigan, M. (2017). Green marketing orientation: Conceptualization, scale development and validation. *Journal of Business Research*, 80(11), 236–246. <https://doi.org/10.1016/j.jbusres.2017.05.024>
- Paul, J., & Benito, G. R. (2018). A review of research on outward foreign direct investment from emerging countries, including China: What do we know, how do we know and where should we be heading? *Asia Pacific Business Review*, 24(1), 90–115. <https://doi.org/10.1080/13602381.2017.1357316>

- Perkins, M., & Roe, J. (2024). Decoding academic integrity policies: A corpus linguistics investigation of AI and other technological threats. *Higher Education Policy*, 37(3), 633–653. <https://doi.org/10.1057/s41307-023-00287-1>
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *The Journal of Applied Psychology*, 88(5), 879–903. <https://doi.org/10.1037/0021-9010.88.5.879>
- Polit, D. F., & Beck, C. T. (2006). The content validity index: Are you sure you know what's being reported? Critique and recommendations. *Research in Nursing & Health*, 29(5), 489–497. <https://doi.org/10.1002/nur.20147>
- Polit, D. F., Beck, C. T., & Owen, S. V. (2007). Is the CVI an acceptable indicator of content validity? Appraisal and recommendations. *Research in Nursing & Health*, 30(4), 459–467. <https://doi.org/10.1002/nur.20199>
- Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences*, 20(9), 676–688. <https://doi.org/10.1016/j.tics.2016.07.002>
- Rodrigues, M., Silva, R., Borges, A. P., Franco, M., & Oliveira, C. (2025). Artificial intelligence: Threat or asset to academic integrity? A bibliometric analysis. *Kybernetes*, 54(5), 2939–2970. <https://doi.org/10.1108/K-10-2023-1329>
- Saihi, A., Ben-Daya, M., & Hariga, M. (2025). The moderating role of technology proficiency and academic discipline in AI-chatbot adoption within higher education: Insights from a PLS-SEM analysis. *Education and Information Technologies*, 30(5), 5843–5881. <https://doi.org/10.1007/s10639-024-13023-0>
- Salah, M., Abdelfattah, F., Alhalbusi, H., & Al Mukhaini, M. (2024). Me and My AI Bot: Exploring the 'aiholi' phenomenon and university students. *Dependency on Generative AI Chatbots-Is This the New Academic Addiction?* Research Square. <https://doi.org/10.21203/rs.3.rs-3508563/v2>
- Sarstedt, M., Hair, J. F., Jr, Cheah, J. H., Becker, J. M., & Ringle, C. M. (2019). How to specify, estimate, and validate higher-order constructs in PLS-SEM. *Australasian Marketing Journal*, 27(3), 197–211. <https://doi.org/10.1016/j.ausmj.2019.05.003>
- Shengelia, R., Gabisonia, L., Tsiklauri-Shengelia, Z., & Shengelia, N. (2024). Assessing AI dependency: A multifactorial analysis and AIDI Index. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4716709>
- Striepe, M., Thomson, S., & Sefcik, L. (2023). Understanding academic integrity education: Case studies from two Australian universities. *Journal of Academic Ethics*, 21(1), 1–17. <https://doi.org/10.1007/s10805-022-09450-9>
- Sultana, A., & Farooq, R. (2025). Towards the measurement of stereotype threat: Scale development and validation. *Global Knowledge, Memory and Communication*, 74(9–10), 3222–3241. <https://doi.org/10.1108/GKMC-10-2023-0378>
- Waltz, C., Lea Strickland, O., & Lenz, E. (2010). *Measurement in nursing and health research*. Springer International Publishing.
- Wang, F., Li, N., Cheung, A. C., & Wong, G. K. (2025). In GenAI we trust: An investigation of university students' reliance on and resistance to generative AI in language learning. *International Journal of Educational Technology in Higher Education*, 22(1), 59. <https://doi.org/10.1186/s41239-025-00547-9>
- Wang, T., Lund, B. D., Marengo, A., Pagano, A., Mannuru, N. R., Teel, Z. A., & Pange, J. (2023). Exploring the potential impact of artificial intelligence (AI) on international students in higher education: Generative AI, chatbots, analytics, and international student success. *Applied Sciences*, 13(11), 6716. <https://doi.org/10.3390/app13116716>
- Worthington, R. L., & Whittaker, T. A. (2006). Scale development research: A content analysis and recommendations for best practices. *The Counseling Psychologist*, 34(6), 806–838. <https://doi.org/10.1177/0011000006288127>
- Yeo, M. A. (2023). Academic integrity in the age of artificial intelligence (AI) authoring apps. *TESOL Journal*, 14(3), e716. <https://doi.org/10.1002/tesj.716>
- Young, K. S. (1998). Internet addiction: The emergence of a new clinical disorder. *CyberPsychology & Behavior*, 1(3), 237–244. <https://doi.org/10.1089/cpb.1998.1.237>
- Yusoff, M. S. B. (2019). ABC of content validation and content validity index calculation. *Education in Medicine Journal*, 11(2), 49–54. <https://doi.org/10.21315/eimj2019.11.2.6>
- Zhang, S., Zhao, X., Zhou, T., & Kim, J. H. (2024). Do you have AI dependency? The roles of academic self-efficacy, academic stress, and performance expectations on problematic AI usage behavior. *International Journal of Educational Technology in Higher Education*, 21(1), 34. <https://doi.org/10.1186/s41239-024-00467-0>
- Zhang, X., Yin, M., Zhang, M., Li, Z., & Li, H. (2025). The development and validation of an Artificial Intelligence Chatbot Dependence Scale. *Cyberpsychology, Behavior and Social Networking*, 28(2), 126–131. <https://doi.org/10.1089/cyber.2023.0175>
- Zunaidah, A., Wiharja, C. K., & Febriantono, M. A. (2023). Dependency on AI-based writing tools in English learning: Implications for human-computer interaction. In *Proceedings of the 2023 International Conference on Information Management and Technology (ICIMTech)*. IEEE. <https://doi.org/10.1109/ICIMTech59029.2023.10278054>

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